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(57) **ABSTRACT**

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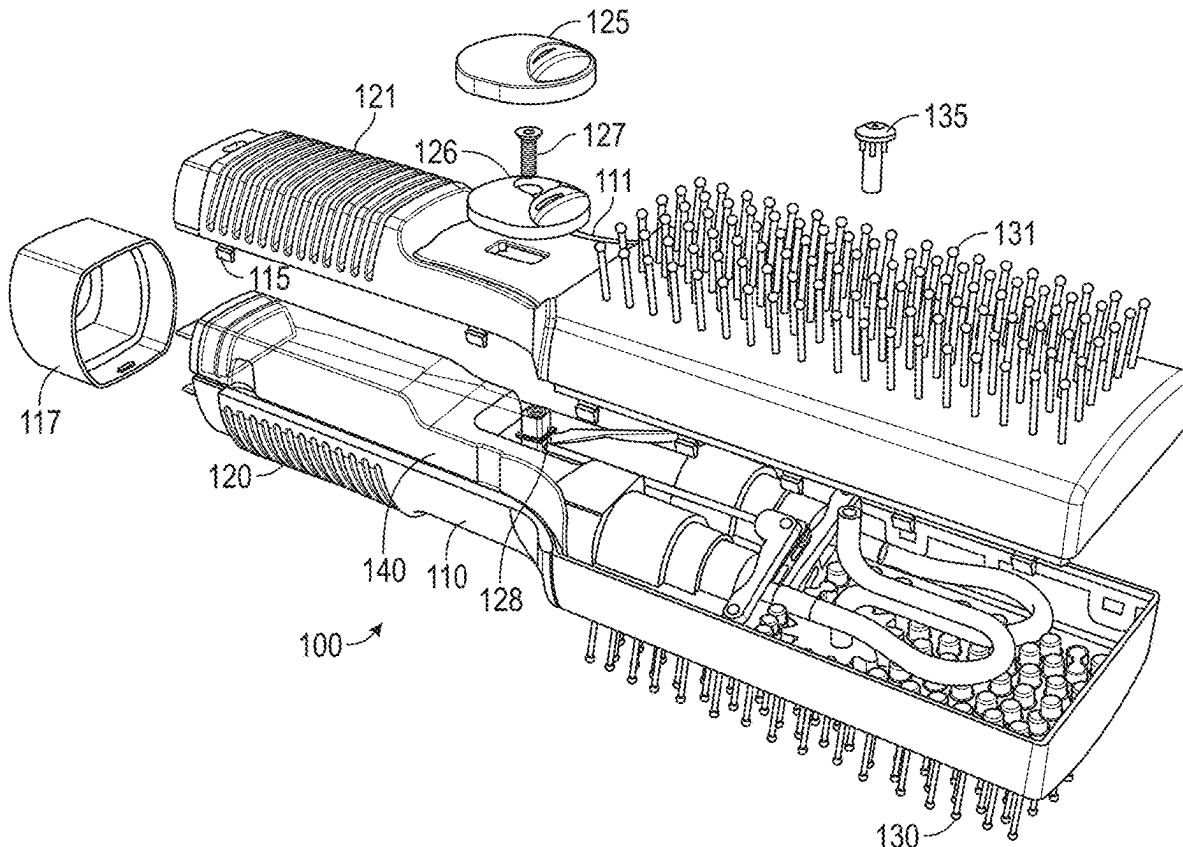
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The present invention is directed to a multi liquid dispensing pump brush, specifically, a hairbrush that is capable of dispensing multiple low viscosity liquids simultaneously. When a consumer desires to detangle and style hair, often times one hair liquid or one hairstyling tool is not sufficient. Thus, hair care can become a cumbersome process as the consumer attempts to maneuver multiple brushes along with multiple containers of liquids. Depending on a consumer's hair needs, one hairbrush that can dispense multiple liquids from multiple surfaces would eliminate the hassle of needing to constantly switch hairstyling tools, and it would enable the consumer to achieve his or her desired hairstyling results through the usage of this one all-encompassing tool.



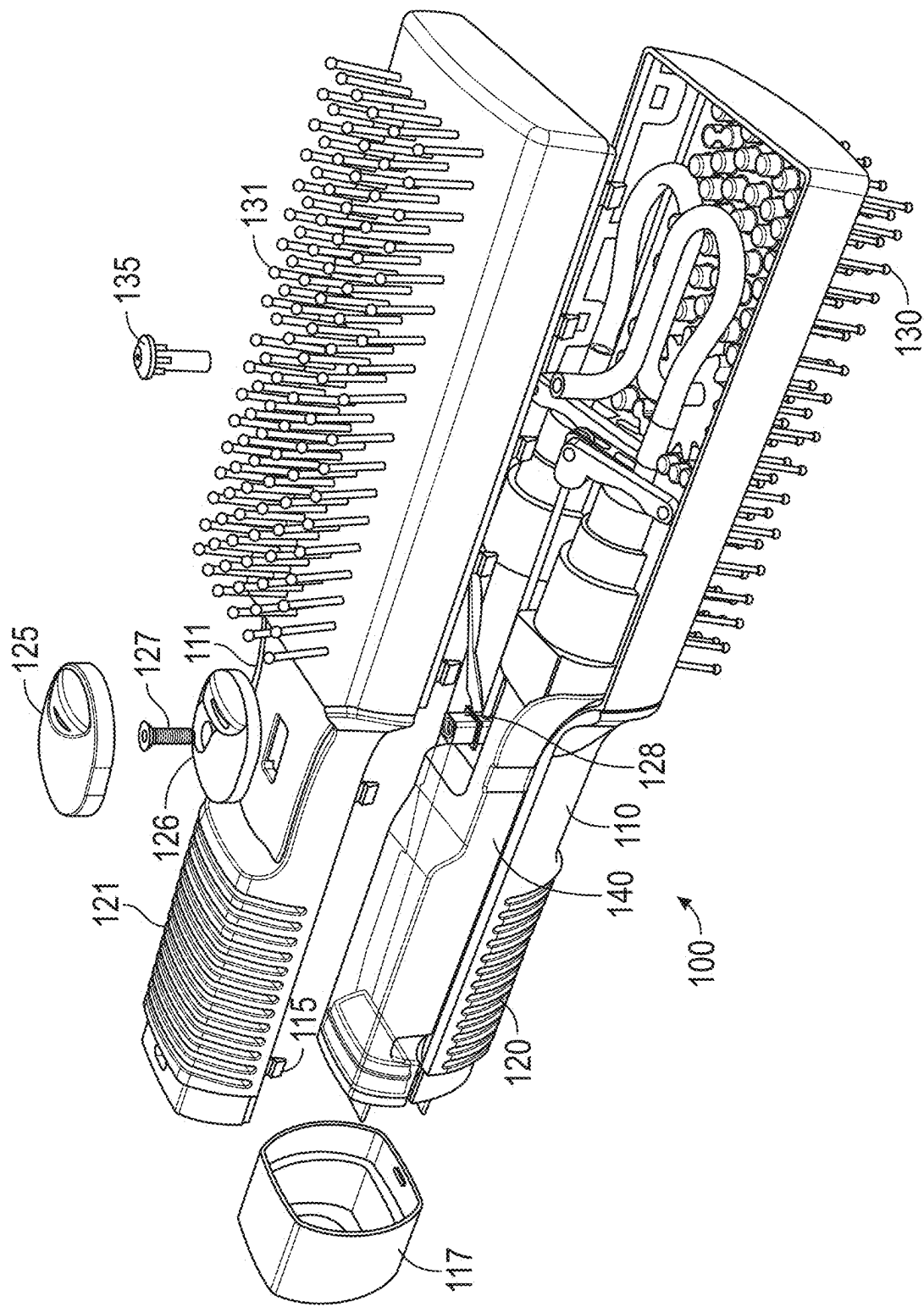


FIG. 1

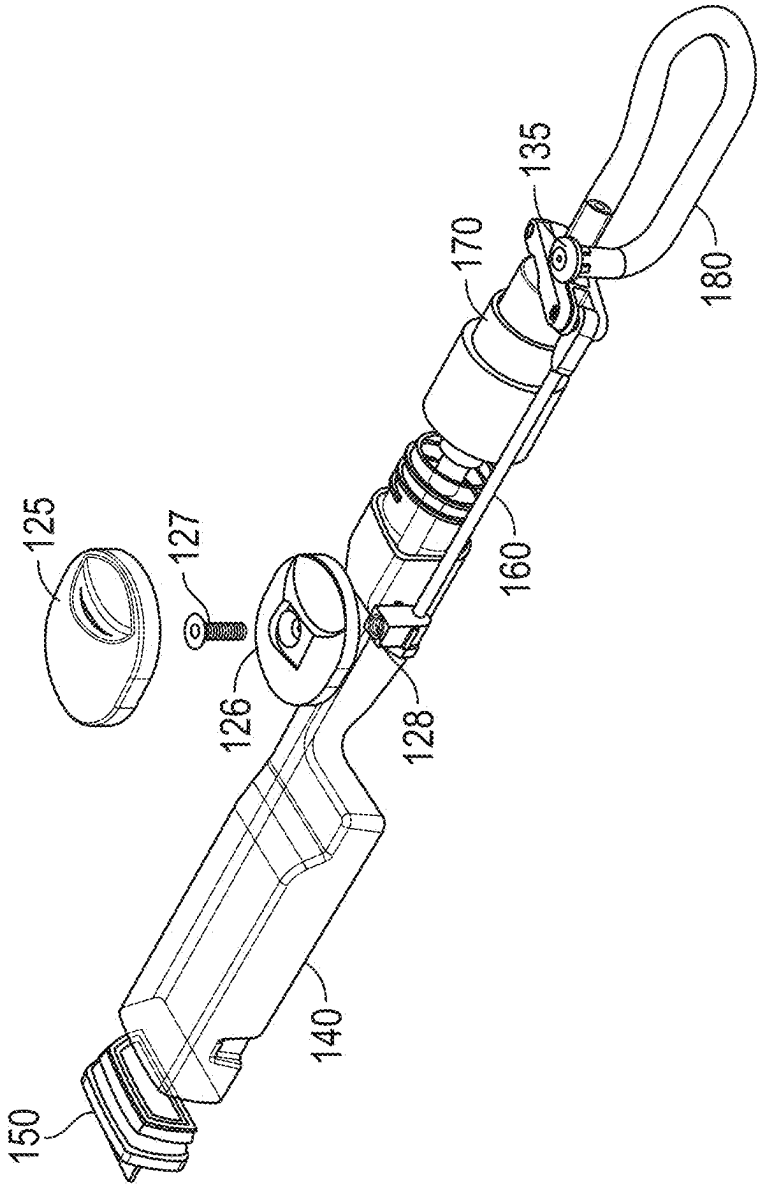


FIG. 2

MULTI LIQUID DISPENSING PUMP BRUSH

CROSS REFERENCES TO RELATED APPLICATIONS

[0001] The present application is a continuation-in-part of U.S. application Ser. No. 17/579,883 filed on Jan. 20, 2022. That application claimed the benefit of U.S. Provisional Patent Application Ser. No. 63/140,120, filed on Jan. 21, 2021.

TECHNICAL FIELD

[0002] The present technology is related to liquid dispensing hairbrushes, systems, and associated devices. In particular, various embodiments of the present technology are related to liquid dispensing hairbrushes with pump dispensers and removable liquid containers.

BACKGROUND OF INVENTION

[0003] The beauty industry, specifically the hair care industry, involves skilled professionals who provide hairstyling services to consumers in public venues such as salons, barbershops and day spas. Outside of these venues, consumers have long been able to purchase commercial hair care products to achieve the same hairstyling goals at home. One such product is a hairbrush, a handle brush with rigid or soft bristles that can be used for smoothing, styling and detangling human hair, animal hair, or animal fur. In addition to the hairbrush, a number of hairstyling liquids (i.e. hair cosmetics including conditioners, styling gels, styling sprays, oils, etc.) are used by consumers at home. Contingent on a consumer's hair texture and desired hairstyling goals, the process of hair care may require several of these liquids. Accordingly, some consumers may find themselves surrounded and overwhelmed by multiple containers of liquids, multiple hairbrushes, and multiple other hairstyling tools, all to engage in a time consuming and laborious process of hair care.

[0004] While there are liquid-dispensing hairbrushes in the prior art that effectively eliminate the need for a separate container of liquid, there are limitations that exist with such hairbrushes. Existing liquid-dispensing hairbrushes do not accommodate consumers that may require several hairstyling liquids; Nor do they accommodate the need for these liquids to be dispensed through varying bristle patterns on a hairbrush, which may be necessary due to hair texture, hair styling, or hair maintenance needs. For liquids having various viscosities, such as low viscosity liquids (i.e. leave in conditioners or styling sprays), the application of such liquids into hair requires different mechanisms for dispensing.

[0005] For consumers with such needs, once again, they may find themselves surrounded and overwhelmed by multiple containers of liquid and/or multiple liquid-dispensing hairbrushes, all to engage in a laborious process of hair care. Therefore, it is advantageous to alleviate this burden by developing a single hairbrush apparatus utilizing a pump mechanism to provide a consumer with the ability to dispense multiple liquids through varying bristle patterns of one hairbrush.

SUMMARY OF INVENTION

[0006] The multi liquid dispensing pump hairbrush of the present invention externally resembles existing hairbrushes,

such that, it has a handle and a brush head with bristles. Internally, this hairbrush further comprises a pump dispenser system, which runs through the longitudinal axis of the hairbrush, consisting of multiple interconnections of liquid containers and tubes. Sliders on the handle are designed to interact with each of these interconnections. As a slider is pumped, or slid, in small up and down motions, it can facilitate the transportation of a liquid through the hairbrush. Ultimately, it can dispense the liquid through dispensing nozzles that are embedded within the bristles on the brush head.

[0007] In the preferred embodiment, externally the hairbrush has two surface areas each comprising a bristle pattern, a pump slider, and a dispensing nozzle. Internally, the hairbrush has two tubes that are attached to two liquid containers on one end and attached to the two dispensing nozzles on the other end. As an initial step, a user fills each liquid container with a low viscosity hairstyling liquid, enabling two different liquids to be transported through this hairbrush. By pumping, or sliding, each of the two pump sliders in small up and down motions, it activates an internal pump dispenser, which transports the liquids from the liquid cartridges, through the tubes, and out of the two dispensing nozzles. It should be noted that the preferred embodiment contains two patterns of bristles providing a user flexibility for hairstyling needs. For example, one preferred detangling liquid can be dispensed from the dispensing nozzle that is embedded within the bristles best patterned for detangling. Similarly, a second preferred styling liquid can be dispensed from the dispensing nozzle that is embedded within the bristles best patterned for styling, enabling a combination of two liquids to be used for hairstyling.

[0008] In other embodiments, the hairbrush can be of any shape and can have any plurality of pump sliders, liquid containers, tubes, nozzles, and bristle patterns.

BRIEF DESCRIPTION

[0009] FIG. 1 is an exploded front view of an external frame of a multi liquid dispensing pump brush.

[0010] FIG. 2 is a perspective view of an internal liquid dispensing system of the multi liquid dispensing pump brush.

DETAILED DESCRIPTION

[0011] FIG. 1 is an exploded front view of an external frame of a multi liquid dispensing pump brush 100 constructed in accordance with the invention. The multi liquid dispensing pump brush 100 is divided into two primary components: an external frame and an internal liquid dispensing system. The external frame comprises: a first frame component 110, a second frame component 111, snap fasteners 115, a handle cover 117, handle grips 120 and 121, brush bristles 130 and 131, pump sliders 125 (only one pump slider is depicted from this view), bases 126 (only one base is depicted from this view), screws 127 (only one screw is depicted from this view), and a dispensing nozzle 135 (only one dispensing nozzle is depicted from this view). In this exploded front view of the external frame, a portion of the internal liquid dispensing system is shown: a slider connector 128 and a liquid cartridge 140. The mechanisms and processes for the internal liquid dispensing system are further described below in FIG. 2.

[0012] In the current embodiment, the first frame component 110 is capable of housing a first portion of the liquid dispensing system, and the second frame component 111 is capable of housing a second portion of the liquid dispensing system. The first and second frame components 110 and 111 are removably attached to one another through the snap fasteners 115. A minimum of seven snap fasteners are used in this embodiment to facilitate a secure attachment, but in other embodiments, any number of snap fasteners or any other means of attachment devices can be used. Once the first and second frame components 110 and 111 are snapped into place, or attached to one another, the external frame of the multi liquid dispensing pump brush 100 is formed. A handle cover 117 is secured on the bottom edge of the handle of the brush. The external frame is designed to be easily attachable and detachable to facilitate the removal, replacement, and refilling of the liquid cartridges 140 housed within.

[0013] The handle grips 120 and 121 can be molded from a firm rubber, or from any similar material commonly known in the art. The current embodiment depicts the handle grips having grooves arranged in parallel lines, which can assist a User with grip strength and comfort while holding the brush. However, it should be noted that the handle grips can be presented in any shape, in any format, or in any manner. In other embodiments, handle grips may be omitted, as well.

[0014] The top portion of the first frame component 110 contains brush bristles 130. The top portion of the second frame component 111 contains brush bristles 131, and further shows the dispensing nozzle 135 embedded within the brush bristles 131. Although not shown in this view, a dispensing nozzle is also embedded within brush bristles 130. The dispensing nozzles 135 are the release points for the liquids contained within the liquid cartridges 140 and are located on both the first and second frame components 110 and 111 (i.e. on both side of the multi liquid dispensing pump brush). Although not shown in this view, at least two liquid cartridges 140 are housed inside of the external frame. These liquid cartridges are each operably connected to the dispensing nozzles. As a result, at least two liquids are capable of being stored inside the brush, transported through the brush, and dispensed out of the brush. The brush bristles 130 and 131 may each have distinct patterns of bristles amongst them to better accommodate the needs of detangling or styling hair. The current embodiment shows one dispensing nozzle 135 within brush bristles 131, but any number of dispensing nozzles may be incorporated into other embodiments of the present invention.

[0015] Further shown on the external side of the second frame component 111 is a pump slider 125, a base 126, and a screw 127, all capable of attaching to a slider connector 128 on the internal side of the second frame component. The screw 127 attaches the pump slider 125 to the base 126 and to the slider connector 128. Although not shown in this view, a pump slider, a base, and a screw are identically positioned on the external side of the first frame component 110, and a slider connector is identically positioned on the internal side of the first frame component. The slider connectors 128 are the connection points between the external frame and the internal liquid dispensing system. As the User pumps, or slides, the pump sliders 125 in small up and down motions, it causes the slider connectors 128 to activate the internal

liquid dispensing system and dispense the liquids contained within liquid cartridges 140. These processes are described in further detail below.

[0016] FIG. 2 is a perspective view of the internal liquid dispensing system of the multi liquid dispensing pump brush. While one liquid cartridge 140 is shown in this figure, the internal liquid dispensing system comprises at least two liquid cartridges, each of which consist of its own slider connector 128, pull rod 160, pump dispenser 170, tube 180, and plug 150.

[0017] In operation, the liquid dispensing system stores, transports, and dispenses liquid through the multi liquid dispensing pump brush as follows: The liquid container 140 is filled with liquid from its bottom edge, and the plug 150 is then inserted to create a closed seal. As the User pumps, or slides, either of the pump sliders 125 in small up and down motions, it activates the slider connector 128, which in turn pulls and pushes the pull rod 160, causing the pump dispenser 170 to pump the liquid out of the liquid cartridge 140 into one end of the tube 180. The pump dispenser is commonly known in the art as a standard push down top pump dispenser. While one end of the tube 180 is attached to the pump dispenser 170, the other end of the tube 180 is attached to the dispenser nozzle 135. Ultimately, the liquid is transported from the liquid cartridge, through the pump dispenser, into the tube, and dispensed through the dispenser nozzle, and into the User's hair.

[0018] At least two liquid cartridges are necessary for the functionality of the multi liquid dispensing pump brush. The liquid cartridges are both housed inside of the brush frame. Thus, they are designed to sizably fit with one another inside of the external frame. The liquid cartridges are removable by a user and can be filled with any liquid of choice. The plug 150 is found at the bottom of the liquid cartridge to provide an open and close mechanism on the cartridge and easy access to the liquid contained within. In its open position, the liquid cartridge 140 can be filled, refilled or cleaned. In its closed position, the plug 150 seals the mouth of the liquid cartridge closed. By sealing this liquid cartridge, a user can be assured that the liquid will not leak from the end of the liquid cartridge, but that it will continue down its path of travel through the multi liquid dispensing pump brush.

[0019] In operation, the User that desires a combination of multiple liquids for the purposes of detangling hair and styling hair, will benefit from the versatility provided by this brush. Using a variety of bristle patterns and multiple liquids, the User can style his or her hair in a more convenient manner. In the current embodiment, the length of the brush is approximately 11 inches, while the width of the brush head is approximately 3 inches. Additionally, the components are constructed from a variety of plastic materials.

[0020] The components of the brush, the liquid containers, and the tubes can be constructed of any material having similar or more proficient properties that are well known to those skilled in the art to be used safely for hairbrushes. The components of this invention can vary in range of height, length, width and weight, as well as, in thickness and in size. Further, it can be made in any preferred color or design, customizable or pre-printed. Although the present invention has been described with reference to specific exemplary embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the invention

as set forth in the claims. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

1. A multi liquid dispensing pump brush comprising:
 - a first frame component, which houses a first liquid container, a first pump dispenser, and a first tube;
 - a second frame component, which houses a second liquid container, a second pump dispenser, and second tube;
 - a first sliding component;
 - a second sliding component;
 - wherein a portion of the first frame component comprises a first bristle pattern and a portion of the second frame component comprises a second bristle pattern;
 - wherein a first aperture is embedded within the first bristle pattern of the first frame component and a second aperture is embedded within the second bristle pattern of the second frame component;
 - wherein the first frame component and the second frame component are removably attached to each other to form an external frame of the multi liquid dispensing pump brush, housing the two liquid containers, the two pump dispensers, and the two tubes within.
2. The multi liquid dispensing pump brush of claim 1, wherein the first liquid container is removable from the brush and fillable with a first liquid and the second liquid container is removable from the brush and fillable with a second liquid.
3. The multi liquid dispensing pump brush of claim 2, wherein the first liquid container and the second liquid container are designed to both fit together inside of the external frame.
4. The multi liquid dispensing pump brush of claim 3, wherein the first sliding component is slidable on the first frame component and the second sliding component is slidable along the second frame component.
5. The multi liquid dispensing pump brush of claim 4, wherein the first sliding component engages a first slider connector, which is capable of pulling and pushing a first pull rod, and the second sliding component engages a second slider connector, which is capable of pulling and pushing a second pull rod.
6. The multi liquid dispensing pump brush of claim 5, wherein the first pull rod is capable of pumping the first

pump dispenser, and the second pull rod is capable of pumping the second pump dispenser.

7. The multi liquid dispensing pump brush of claim 6, wherein the first tube is connected on one end to the first pump dispenser and on the other end to the first aperture, and the second tube is connected on one end to the second pump dispenser and on the other end to the second aperture.

8. The multi liquid dispensing pump brush of claim 7, wherein each sliding component engages with its own pipeline consisting of the slider connector, the liquid container, the pump dispenser, the tube, and the aperture.

9. The multi liquid dispensing pump brush of claim 8, wherein the first sliding component slides along the first frame component, drives the transfer of the first liquid through the first liquid container, through the first pump dispenser, through the first tube, and through the aperture on the first frame component, and the second sliding component slides along the second frame component, drives the transfer of the second liquid through the second liquid container, through the second pump dispenser, through the second tube, and through the aperture on the second frame component.

10. The multi liquid dispensing pump brush of claim 4, wherein the first sliding component is manually operable to drive the first liquid through the aperture of the first frame component, and the second sliding component is manually operable to drive the second liquid through the aperture of the second frame component.

11. The multi liquid dispensing pump brush of claim 1, wherein the first bristle pattern and the second bristle pattern are distinctly different patterns.

12. The multi liquid dispensing brush of claim 1, wherein the multi liquid dispensing brush is composed of plastic materials.

13. The multi liquid dispensing pump brush of claim 1, wherein the first frame component and the second frame component are attached to one another using snap fasteners.

14. The multi liquid dispensing brush of claim 1, wherein each liquid container further comprises:

- a body having a mouth,
- a plug used to seal the mouth of the container,
- wherein the body defines a chamber configured to contain a liquid.

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